GLUCOSE DIGITAL TWIN — END-TO-END SYSTEM ARCHITECTURE

Physiological Continuous Glucose Forecasting & Participant-Specific Residual Calibration Engine

STAGE 01

DATA INGESTION & QC

PhysioNet CGMacros (N=45)

Cohort: 45 subjects (15 Normal, 16 Prediabetes, 14 T2D).
Contiguous Runs: Delta t = 60s validation.
Zero Gap Imputation: Runs end on missing data.
Lattice Grid: m(t) = round((t-t0)/60s), issuance on m(t) mod 5 == 0.

SAFEGUARD: Zero boundary bleed; strict 1-min raw validation.

STAGE 02

GUARDED FEATURES

R^20 Multimodal Matrix

History (13 lags): y(t-65) through y(t-5).
5-Min Gap: Zero input permitted in [t-5, t].
Meal Timing (3): Minutes since meal, meal in 60m, no recent meal.
Clinical Demographics (4): Age, Gender, BMI, baseline HbA1c.

SAFEGUARD: 5-min guard band prevents interpolation leakage.

STAGE 03

GLOBAL BENCHMARK

Disjoint CV Ladder (Rungs 0-6)

Disjoint CV: 10 repeats x 5 folds (50 folds, K=45).
Train-Only Tuning: Closed-form Ridge alphas.
Persistence: RMSE 21.55 (+30m), 29.32 (+60m).
Rung 5 Ridge: RMSE 18.85 / 25.45 (+11.8% Skill).
Rung 6 GBDT: Parity with Ridge (spans zero).

SAFEGUARD: 0% intra-subject identity bleed.

STAGE 04

PERSONALIZATION

Stage 3 Residual Calibration

Adaptation Days: Early 2, 4, 6 study days.
Residual Model: Ridge on r(t) = y(t+h) - y_hat.
6-Day Impact: -1.81 mg/dL (-8.0% error, p < 0.05).
Cohort Gain: 40/45 improved (88.9%).
Streamlit UI: 4 interactive research modules.

SAFEGUARD: >= 120-min purge buffer prevents lag spillover.

BENCHMARK VERIFICATION GUARANTEES: 123,599 valid origins across 45 participants | 50 disjoint folds | Global Skill: +11.82% (+30m), +11.85% (+60m) vs Persistence | 6-Day Personalization: -1.81 mg/dL (-8.0% error, p < 0.05, 40/45 improved) | Zero data leakage